

IPC144 – Introduction to C

Programming

Practice Exercises

Week 4

Formatted Input / Output

9.7 (Differences Between %d and %i) Write a program to test the difference between the %d and %i conversion specifiers when used in scanf statements. Ask the user to enter two integers separated by a space. Use the statements

```
scanf("%i%d", &x, &y);  
printf("%d %d\n", x, y);
```

to input and print the values. Test the program with the following sets of input data:

```
10 10  
-10 -10  
010 010  
0x10 0x10
```

9.8 (Printing Numbers in Various Field Widths) Write a program to test the results of printing the integer value 12345 and the floating-point value 1.2345 in fields of various sizes. What happens when the values are printed in fields containing fewer digits than the values?

9.9 (Rounding Floating-Point Numbers) Write a program that prints the value 100.453627 rounded to the nearest digit, tenth, hundredth, thousandth and ten-thousandth.

9.10 (Temperature Conversions) Write a program that converts integer Fahrenheit temperatures from 0 to 212 degrees to floating-point Celsius temperatures with 3 digits of precision. Perform the calculation using the formula

$$\text{celsius} = 5.0 / 9.0 * (\text{fahrenheit} - 32);$$

The output should be printed in two right-justified columns of 10 characters each, and the Celsius temperatures should be preceded by a sign for both positive and negative values.

9.11 (Escape Sequences) Write a program to test the escape sequences '\', '\n', '\t', '\a', '\b', '\f', '\r', and '\t'. For the escape sequences that move the cursor, print a character before and after printing the escape sequence so it's clear where the cursor has moved.

9.12 (*Printing a Question Mark*) Write a program that determines whether ? can be printed as part of a printf format control string as a literal character rather than using the \? escape sequence.

9.13 (*Reading an Integer with Each scanf Conversion Specifier*) Write a program that inputs the value 437 using each of the scanf integer conversion specifiers. Print each input value using all the integer conversion specifiers.

9.14 (*Outputting a Number with the Floating-Point Conversion Specifiers*) Write a program that uses each of the conversion specifiers e, f and g to input the value 1.2345. Print the values of each variable to prove that each conversion specifier can be used to input this same value.

9.15 (*Reading Strings in Quotes*) In some programming languages, strings are entered surrounded by either single or double quotation marks. Write a program that reads the three strings suzy, "suzy" and 'suzy'. Are the single and double quotes ignored by C or read as part of the string?

9.16 (*Printing a Question Mark as a Character Constant*) Write a program that determines whether ? can be printed as the character constant '?' rather than the character constant escape sequence '\?' using conversion specifier %c in the format control string of a printf statement.

9.17 (*Using %g with Various Precisions*) Write a program that uses the conversion specifier g to output the value 9876.12345. Print the value with precisions ranging from 1 to 9.